

CHAPTER

36



Sthaulya Karshya

Learning Objectives

- ⇒ To learn the concept of Sthaulya
- ⇒ To learn the Nidana and Samprapti of Sthaulya
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To describe Samanya and Vishesha Lakshana of Sthaulya
- ⇒ To know about Samanya Lakshana of Karshya
- ⇒ To know about the differential diagnosis of Karshya

Introduction

Sthaulya¹ is *Medo Prajoshaja Vikara*, where we can observe excess abnormal *Medo Dhatu* in the individual. As per *Madhavakara*, it is also known as *Medoroga* (a condition where *Medo Dhatu* is excessively and abnormally increased). As per *Charaka*, it is categorised under *Santarpanaja Vikara* i.e. disease caused by over nutrition of the body. This is more commonly seen in the people who are having sedentary lifestyle. In other words to say, when an individual is having more input or intake than output or work, then the person will have excess deposition of *Medas* (fat) in different parts of the body. When one analyses the *Dosha* vitiation, this is manifested due to the increase of *Kapha Dosha* inside the body. As far as the type of the disease is concerned, this is categorised in *Sharirika* (somatic disease) category of *Vikara* under the subheading of *Doshabala Pravritta Vikara* which is categorised under

Adhyatmika Vikara. This is having *Udbhava Sthana* as *Amashaya*, *Sanchara Sthana* as *Sarva Sharira* (whole body) and *Vyakta Sthana* as *Tvak* (skin), *Sphik* (buttocks), *Udara* (abdomen) and *Stana* (breasts).

Definition of Atisthula

मेदोमांसातिवृद्धत्वाच्चलस्फिगुदरस्तनः।
अयथोपचयोत्साही नरोऽतिस्थूल उच्यते॥१॥

Ca.Su. - 21/9

The clinical features of *Atisthula* are

- ⊗ Pendulous abdomen
- ⊗ Pendulous buttocks
- ⊗ Pendulous breasts²
- ⊗ Disproportionate increase in the size of the individual³
- ⊗ Poor enthusiasm in any activity⁴

- 1 Sthaulya is a clinical condition where in the person will have excess of body fat. This results mainly due to the disproportionate energy intake and energy expenditure, where intake is much higher than expenditure. When an individual consumes more amount of food either carbohydrates, fats or proteins, that will lead to the conversion of respective micromolecules to fat and these converted materials will move to adipose tissues and stored inside the adipose tissues as great as 85-90% of the total adipose tissues.
- 2 These features are attributable to the disproportionately increased deposition of adipose tissue in the abdomen, buttocks and breasts beneath the skin and these are the least used parts of the body when one considers the overall activity of the human body.
- 3 When an individual is consuming excess amount of food be carbohydrate or fat or protein, will be stored entirely in the form of adipose tissue, to be used for energy later in due course of time. Adipocytes are modified fibroblasts that store almost pure triglycerides in quantities as great as 80 to 95% of the cell volume. These will be stored in subcutaneous tissues in different parts of the body such as abdomen, buttocks and breasts etc. Thus, the excess accumulation of fat tissue more than normal results in the disproportionate increase in the size of the individual.
- 4 Regular physical exercise will lead to the increase of muscle mass and decrease of fat mass of the body, where as lack of physical exercise will lead to increase of fat mass and decrease of muscle mass in the body. Excess accumulation of fat in different parts of the body more

❖ Excess Medas⁵ and Mamsa in the body



Figure 159: Sthaulya

Nidana of Sthaulya

तदतिस्थौल्यमतिसंपूरणाहरुमधुरशीत स्निग्धोपयोगादव्यायामादव्यवा-
यादिवास्वप्नाद्धर्षनित्यत्वादचिन्तनाद्वीजस्वभावाश्चोपजायते।

तस्य यतिमात्रमेदै-

Ca.Su. – 21/4

अव्यायामदिवास्वप्न श्लेष्मलाहारसेविनः।

मधुरोऽन्नरसः प्रायः स्नेहा मेदः प्रवर्धयेत्॥1॥

Ma.Ni. – 34/1

संतर्पयति यः स्निग्धैर्मधुरैर्गुरुपिच्छिलैः।

नवान्नैर्नवमद्यैश्च मांसैश्चानूपवारिजैः॥3॥

गोरसैर्गडिकैश्चानैः पैष्टिकैश्चातिमात्रशः।

चेष्टाद्वेषी दिवास्वप्नशय्यासनसुखे रतः॥4॥

रोगास्तस्योपजायन्ते संतर्पणनिमित्तजाः।

Ca.Su. – 23/3-5

- ❖ Excess⁶ intake of food substances
- ❖ Excess intake of Guru (heavy to digest), Madhura (sweet taste), Snigdha (unctuous) and Shita (cold in touch and potency) substances⁷
- ❖ Physical inactivity⁸
- ❖ Less indulgence in sexual intercourse⁹
- ❖ Sleeping during day time¹⁰
- ❖ Not having any kind of physical or mental stress¹¹
- ❖ Not having thinking¹²
- ❖ Through genetics¹³
- ❖ Excess intake of substances which provoke Kapha Dosh¹⁴
- ❖ Excess intake of Picchila (slimy) food substances¹⁵
- ❖ Excess intake of freshly harvested grains
- ❖ Excess intake of freshly prepared alcohol
- ❖ Excess intake of meat of marshy and aquatic animals¹⁶
- ❖ Excess intake of milk and milk products¹⁷
- ❖ Excess intake of substances prepared out of flour
- ❖ Activities of sedentary lifestyle¹⁸

Samprapti of Sthaulya

मेदसोवृत्तमार्गत्वात् पुष्यन्त्यन्ये न धातवः।

मेदस्तु चीयते तस्मादशक्तः सर्वकर्मसु॥2॥

Ma.Ni. – 34/2

❖ Medo Dhatu is abnormally increased and no other Dhatu is nourished

❖ The person cannot perform any activities because of this abnormal increase

than excess will lead to sluggish activity of the body parts. He may have laziness secondary excess fat deposits in the body.

5 Excess accumulation of Medas (fat) occurs just beneath the skin i.e. subcutaneous tissue all over the body where we can see the more soft curvatures on the skin for example, in the wrist joint etc.

6 The consumption of food in excess in greater amounts than the body can utilise the same will lead to accumulation of fat. The excess food can be either carbohydrates, fat and proteins can entirely converted to fat in the adipose tissue which can be utilised for energy in future.

7 Excess intake of substances which are Guru such as blackgram items or curds; Madhura taste such as milk, banana or grapes etc; Snigdha such as excess coconut oil or ghee etc; Shita substances such as excess ice cream intake etc

8 In less physical activity, the person will have minimum BMR (basal metabolic rate) and accounts for 50 to 70% of the daily energy expenditure in most sedentary individuals. BMR normally averages about 65-70 Calories per hour in an average 70 kg man.

9 Less indulgence in sexual intercourse will have reduced basal metabolic rate. When there is less metabolism and he is taking the substances in excess or even in normal quantity, may have more accumulation of fat in the adipose tissue.

10 In case of day time sleeping by an individual, there will be obstructive sleep apnoea due to the fall in the tone of the airway dilator muscles leading to the collapse of the upper airways during sleep in day time. This is specifically seen in the patients of obesity.

11 We have seen the effect of lack of the physical stress in the context of physical inactivity. When we speak of mental stress, the person likely to take in excess food may be due to death of a parent or severe illness or mental depression.

12 In this particular situation, the person is not at all having any kind of thinking. The person may have excess consumption of food secondary to lack of mental stress which will be deposited in the body.

13 In this particular situation, the members of the family share many of the same eating habits and physical activity patterns.

14 Consumption of food substances such as excess consumption of substances which are heavy to digest or the substances which are prepared from ghee or curds or sweet substances etc Refer volume 1 written by the author in the context of Kapha Prakopaka Karana.

15 Consumption of food substances which are slimy in nature such as foods prepared by ghee or sugar or jaggery or milk etc

16 Excess intake of meat of fish or pig etc

17 Excess intake of milk, curds, butter milk or ghee etc

18 Activities of sedentary lifestyle include the persons who are having clerical job or the people who are sitting in jewellery shop etc, in toto, all above said aetiological factors will lead to the very high input than its expenditure i.e. supply is more than the demand.

Samprapti Ghataka of Sthaulya

Dosha: Kapha

Dushya: Medas

Srotas: Medovaha Srotas

Srotodushti Prakara: Sanga and Vimargagamana

Agni: Jatharagnimandya and Medo Dhatvagnimandya

Ama : Jatharagnijanya and Medo Dhatvagnijanya

Udbhava Sthana: Amashaya

Sanchara Sthana: Sarva Sharira

Vyakta Sthana : Tvak, Sphik, Udara and Stana

Rogamarga : Abhyantara, Bahya

Vyadhi Svabhava: Chirakari

Sadhyasadyata: Krichrasadya to Asadya

Purvarupa of Sthaulya: No specific Purvarupa are mentioned in Ayurveda literatures for this disease.

Lakshana of Sthaulya

भवन्ति। अतिस्थूलस्य तावदायुषो हासो जवोपरोधः कृच्छ्रव्यवायता दौर्बल्यं दौर्गन्ध्यं खेदाबाधः क्षुदतिमात्रं पिपासातियोगश्चेति भवन्त्यष्टौ दोषाः।

Ca.Su. – 21/4

क्षुद्रश्वास तृषा मोहस्वप्नप्रथनसादनैः।

युक्तः क्षुत्स्वेददुर्गन्धैरल्पप्राणोऽल्पमैथुनः॥३॥

मेदस्तु सर्वभूतानामुदरेष्वस्थिषु स्थितम्।

अत एवोदरे वृद्धिः प्रायो मेदस्थिनो भवेत्॥४॥

Ma.Ni. – 34/3,4

• Decreased life span¹⁹

• Decreased speed in any of the activity taken up by Sthula individual²⁰

- Difficulty in sexual intercourse²¹
- Debility²²
- Bad body odour²³
- Excessive sweating²⁴
- Excessive thirst²⁵
- Excessive hunger
- Breathing discomfort on doing slightest physical exertion²⁶
- Confusion
- Excess sleep
- Sluggish movements of body parts²⁷
- Excess accumulation of Medas in the Udara (abdomen) region²⁸

Introduction

Karshya is a disease involving Medovaha Srotas, where there will be abnormal decrease in Medo Dhatu is observed. As per Ayurveda literatures, Karshya is considered to be Apatarpanajanya Vikara i.e. the disease manifested due to lack of nutrition of the body. In the current era, we are using the scale of BMI²⁹ to determine undernourishment and overnourishment. When the BMI is <18, then we can diagnose that the person is having under nourishment and when the BMI is > 25, then we can diagnose that the person is having over nourishment; when the BMI is > 30, then we can confirm that the person is suffering obesity and when the BMI > 40, then hyperobesity is diagnosed.

Definition of Karshya

शुष्क स्फिगुदरग्रीवो धमनीजालसन्ततः।

त्वगस्थिशोषोऽतिकृशः स्थूलपर्वा नरो मतः॥१५॥

Ca.Su. – 21/15

- G K**
- 19 When an individual is suffering from obesity, he will be having the either more accumulation of fat tissues into the cell and there will be increased circulation of fat tissues in the circulatory system. This pathological mechanism may lead to the manifestation of peripheral arterial disease or myocardial infarction etc complication. When the individual is having excess amount of fat in the circulatory system and if it involves the circulatory system and thus, the person may have decreased life span in case of affliction of heart or peripheral arterial system.
 - 20 This is a vicious cycle. When the person consumes the food which has richness in fat content, that will lead to the excess accumulation of fat in the fat cells. When an individual has less physical activity (sedentary lifestyle), will lead to sluggish movement of the body parts as the person who will have sluggish movement of body parts (people working in jewelry shops, groceries shops etc).
 - 21 This symptom is attributable to the inability of performing the sexual intercourse, as the excess accumulation of fat occurs in the adipose tissue in the subcutaneous tissue, peritoneal cavity in particular. As the size of the abdomen increases in males and females, the person cannot perform the sexual intercourse.
 - 22 It is being said that if an individual performs a single episode of strenuous exercise may increase the basal energy expenditure for several hours after the physical activity is topped. Thus, obese individual performs the strenuous exercise in a single setting will have rapid basal metabolic rate and thus, the person
 - 23 This symptom is attributable to excess sweating. Bad odour is observed due to the presence of urea in the sweat. Sweat contains twice the proportion of urea than that of plasma and this could be the reason for bad body odour.
 - 24 Normally, 400 ml of sweat is produced in 24 hours on an average. When a person is performing severe form of physical activity, this will increase the basal metabolic rate for several hours. This mechanism will lead to increase loss of heat via sweat.
 - 25 When an individual is having increased heat loss, there will be the association of enhanced fluid loss and the presence of thirst indicates hypovolemia in the body. There will be production of 1 liter of sweat per hour in a unacclimatised person. When the person is exposed to hot weather for 1 to 6 weeks, production of sweat increases to as much as 2 to 3 liters per hour. Thus, the person frequently develops thirst owing to upsurge in sweating.
 - 26 This is attributable to the sedentary lifestyle.
 - 27 This is attributable to the excess accumulation of fat in the subcutaneous tissues, buttocks, breast and abdominal region.
 - 28 Fat is stored in adipocytes of subcutaneous tissues and in intraperitoneal cavity. It can also be stored in liver and other tissues of the body.
 - 29 The formula to calculate BMI is $w (kg) / h^2 (m)$. Depending upon the height and weight of the individual, BMI is calculated.

सेवा रूक्षान्नपानानां लङ्घनं प्रमिताशनम्।
क्रियातियोगः शोकश्च वेगनिद्राविनिग्रहः॥1 1॥
रूक्षस्योद्वर्तनं स्नानस्याभ्यासः प्रकृतिर्जरा।
विकारानुशयः क्रोधः कुर्वन्त्यतिकृशं नरम्॥1 2॥

- Excess consumption of Ruksha Annapana (dry food and drinks) or the substances which has less nutritive value³⁰
- Excess fasting or abstinence from food
- Excess consumption of very little amount of food
- Excess exposure to purification therapy such as Panchakarma procedures³¹
- Excess grief³²
- Suppression of natural urges
- Suppression of sleep
- Undergoing Udvartana line of treatment by emaciated person³³
- Frequent bathing³⁴
- As a constitution, person will be lean³⁵
- Ageing³⁶
- As a complication of other chronic debilitating

illnesses³⁷• Anger³⁸

Lakshana of Karshya

व्यायाममत्तिसौहित्यं क्षुत्पिपासामयौषधम्।
कृशो न सहते तद्वदतिशीतोष्णमैथुनम्॥1 3॥
प्लीहा कासः क्षयः श्वासो गुल्मोऽशस्युदराणि च।
कृशं प्रायोऽभिधावन्ति रोगाश्च ग्रहणीगताः॥1 4॥

Ca.Su. – 21/13,14

- The person cannot tolerate exertion caused by excess exercise
- The person cannot tolerate hunger
- The person cannot tolerate thirst³⁹
- The person cannot tolerate strong medicines
- Intolerance to excess cold atmosphere
- Intolerance to excess hot atmosphere⁴⁰
- Intolerance to sexual intercourse⁴¹
- Enlargement of spleen⁴²
- Cough
- Breathing discomfort
- Depletion of body tissues⁴³

G
K

- 30 When an individual consumes Ruksha substances, that will lead to the increase of Ruksha (dry) quality in the body. Ruksha quality is considered to be one of the major qualities of Vata Dosha. Thus, excess consumption will lead to the increase of Ruksha quality of Vata Dosha.
- 31 These aetiological factors will lead to the sequential depletion of carbohydrates, fats and proteins. Glucose is stored as much glycogen as they can – an amount sufficient to supply the energy needs of the body for only 12 to 24 hours. After this, there will be energy utilisation by the body through fat metabolism and this will be followed by protein metabolism. About 60% of the amino acids in the body proteins can easily be converted into carbohydrates and another 40% have chemical configuration that make the conversion into carbohydrates difficult.
- 32 Normally, gastric juice is secreted during interdigestive period in between the meals. This gastric juice is nonoxynitic type or mucus only. When an individual is under emotional stimuli, that will lead to the increase of interdigestive secretion around 50 ml or more per hour. Emaciation of the body occurs in the long run of the course of the illness.
- 33 It is mentioned in Ayurveda literatures, Udvartana line of the treatment is responsible for the Medas and Kaphahara effect in the body. When an individual is exposed to Udvartana more frequently, that may lead to the rapid depletion of the subcutaneous fat in the body.
- 34 As per fundamentals of Ayurveda, frequent bathing is a kind of Svedana therapy. Frequent hot water bathing is responsible for the peripheral vasodilatation and the person develop increased sweating from the body as well as fat mobilisation. This may lead to the shrinkage of intravascular compartment.
- 35 People with Vataja constitution, are usually having lean physique. Ectomorphic individuals usually will have lean body.
- 36 In the process of aging, the person may have nutritional deficiency – sequential loss of carbohydrate, fats and proteins can be observed in due course of time.
- 37 As per Ayurveda fundamentals, Vata Dosha will be abnormally vitiated in chronic debilitating illnesses. Emaciation is one of the clinical features observed in chronic illness. This can be attributable to sequential loss of carbohydrates, fats and proteins. Muscle wasting is a common sequel of nearly all acute and many chronic diseases.
- 38 When an individual is having excess anger, that will lead to the activation of sympathetic nervous system. When this autonomous nervous system is activated, that will lead to the tachycardia, fight and flight response etc. It will lead to the increased basal metabolic rate (BMR) and thus, the person may develop emaciation of the body.
- 39 When the person is having emaciation, that will be associated with profound loss of carbohydrate, fats and proteins. When the person performs exercise, there is no energy reserve to supplement the energy loss owing to decrease in adipose tissue in the subcutaneous tissue, abdomen, breasts and buttocks in particular. That is the reason, the person will have intolerance to exercise, hunger and thirst.
- 40 As the person is having deficient fat reserve from the subcutaneous tissue, the person is not having the ability to withstand the extremes of the temperature variation.
- 41 When the individual indulges in sexual intercourse, it increases basal metabolic rate. As there is shortage of fat in the body, the person may not have sufficient replenishment of energy loss. This may be the reason for the intolerance to sexual intercourse.
- 42 This Lakshana (clinical feature) is observed in case of Medas Kshaya. When there is rapid depletion of adipose tissue, the person may have fatty degeneration of the liver. This will lead to the cirrhosis of the liver, which in turn leads to the portal hypertension and splenomegaly is the outcome of portal hypertension.
- 43 Depletion of the adipose tissues is seen in subcutaneous tissue, abdomen, breasts and buttocks.

- Gulma (chronic localised abdominal swelling⁴⁴)
- Arshas (hemorrhoids)⁴⁵

- Udara (generalised abdominal swelling⁴⁶)
- Grahani Roga (disease afflicting Grahani)

Key summary of the chapter

मेदोमांसा तिवृद्धत्वाचलस्फिगुदरस्तनः। अवथोपचयोत्साहो नरोऽतिस्थूल उच्यते॥१॥

Ca.Su. - 21/9

- The person will have pendulous movement of buttocks, abdomen and breasts because of the excess accumulation of fats in these areas. Because of the excess accumulation, the person is unable to do any of the activities.
- Atisthula Karana are 8 in number. They are Atisampurana; Guru, Shita, Madhura, Snigdha Upayoga; Avyavama; Avyavaya; Achintana; Harshanityatva; Bijasvabhava
- Atisthula Dosha Lakshana are 8 in number. They are Ayusho Hrasa; Javoparodha; Krichravayavayata, Dorgandhya, Daurbalya; Svedabadha, Kshudatimatra and Pipasa Atiyoga.

शुष्कस्फिगुदरग्रीवो धमनीजालसन्ततः। त्वगस्थिशेषोऽतिकृशः स्थूलपर्वो नरो मतः॥१॥५॥

Ca.Su. - 21/15

- Atikarshya person is having atrophied buttocks, abdomen and neck; have abundant visible veins in the body; Tvak and Asthi are seen in the bod and large visible joints.
- Atikarshya Nidana mentioned in Charaka Samhita are Rukshanna Pana Seva; Langhana, Pramitashana; Kriya Atiyoga, Bhi, Shoka; Vega Vinigraha; Nidra Vinigraha; Rukshasya Udvartana; Snanasya Abhyasa; Prakriti and Jara are general Nidana mentioned which may be responsible for the manifestation of Karshya.
- Karshya Lakshana as mentioned by Charaka are inability to tolerate excess amount of Vyayama, Kshut, Pipasa, Amaya, Aushadha; inability to withstand aftereffects of excess coldness, warmth, sexual intercourse; splenic enlargement; cough; depleted body tissues, breathing discomfort, abdominal swelling; disease afflicting Grahani.

Self-Assessment: Questions for Reviews

Long essays

- Enumerate Ashta Nindita Purusha and explain Explain Atisthula and Atikrisha Purusha in detail
- Define Atisthula and Atikrisha Purusha, explain Atisthula Karana and Atisthula Dosha and Karshya Nidana, Lakshana in detail

Short essays

- Define Atisthula and explain Atisthulya Karana and Lakshana
- Explain Medoroga in detail
- Define Karshya. Explain the Nidana and Lakshana of Karshya.
- Explain the Nidana, Samprapti and Lakshana of Karshya
- What are the Nidana and Lakshana of Karshya

MCQs

1. Karshya is

- a. Mamsavaha Sroto Vikara;
- b. Medovaha Sroto Vikara;
- c. Shukravaha Srotovikara;
- d. Raktavaha Srotovikara

Answer: (b)

2. Which of the following is not considered to be Upadrava of Karshya?

- a. Grahani;
- b. Arshas;

- c. Pliha;
- d. Prameha

Answer: (d)

3. Frequent Udvartana, a line of treatment will lead to

- a. Kaphahara;
- b. Medohara;
- c. Both a and b;
- d. None of the above

Answer: (c)

4. Which of the following is Apatarpanajanya Vikara?

- a. Sthaulya;
- b. Karshya;
- c. Prameha;
- d. Prameha Pidaka

Answer: (b)

5. What should be the value of BMI (Body Mass Index) to diagnose undernourishment?

- a. 22 and below;
- b. 15 and below;
- c. 18 and below;
- d. 26 and below

Answer: (c)

6. Atisthula Dosha arein number.

- a. 7;
- b. 8;
- c. 4;
- d. 2

Answer: (b)

7. As per Charaka, Atisthula Karana arein number.

- a. 7;
- b. 8;
- c. 4;
- d. Innumerable

Answer: (b)

- 44 This is attributable to malignant tumour of the abdomen. Cancer cachexia is the sequel in the patients of malignant diseases.
- 45 This is attributable to the consequence of portal hypertension where backflow of the blood in the superior mesenteric vein is present.
- 46 This is attributable to portal hypertension, where ascites, accumulation of fluid in peritoneal cavity is observed.

8. The term Medoroga is coined by

- a. Charaka; b. Sushruta;
c. Vagbhata; d. Madhavakara

Answer: (d)

9. As per Sushruta, Sthaulya is categorised under

- a. Adhyatmika Vikara;
b. Adhibhautika Vikara;
c. Adhidaivika Vikara;
d. Both a and b

Answer: (a)

10. As per Sushruta, Sthaulya is categorised under

- a. Adibala Pravritta Vikara;
b. Janmabala Pravritta Vikara;
c. Doshabala Pravritta Vikara;
d. Daivabala Pravritta Vikara

Answer: (c)

Chapter resources

- Charaka Samhita, Sutra Sthana – 21st chapter, Ashtou Ninditiya Adhyaya

- Madhava Nidana, 34th chapter – Medoroga Nidana Adhyaya
- Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
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- Savill's system of clinical medicine revised and edited by E C Warner, 14th edition, CBS Publishers and Distributors, New Delhi, 1998

